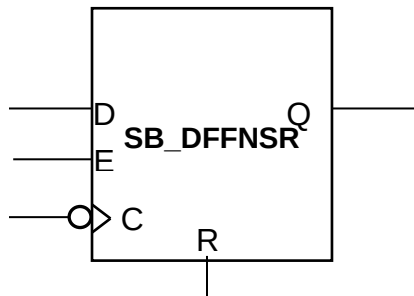


SB_DFFNSR

D Flip-Flop – Negative Edge Clock with Synchronous Reset

Data: D is loaded into the flip-flop when R is low during the falling clock edge transition.

Reset: R input is active high, overrides all other inputs and resets the Q output during the falling clock edge transition.



Inputs			Output
R	D	C	Q
1	X		0
X	X		No Change
0	0		0
0	1		1
Power on State	X	X	0

Key
 Falling Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'
Input C: Logic '0'
Input R: Logic '0'

Verilog Instantiation

```
// SB_DFFNSR - D Flip-Flop – Negative Edge Clock, Reset is synchronous with the falling clock edge
SB_DFFNSR    SB_DFFNSR_inst (
    .Q(Q),           // Registered Output
    .C(C),           // Clock
    .D(D),           // Data
    .R(R)            // Synchronous Reset
);
```

```
// End of SB_DFFNSR instantiation
```

VHDL Instantiation

```
-- SB_DFFNSR - D Flip-Flop – Negative Edge Clock, Reset is synchronous with the falling clock edge
```

```
SB_DFFNSR_inst: SB_DFFNSR
  port map (
    Q => Q,          -- Registered Output
    C => C,          -- Clock
    D => D,          -- Data
    R => R            -- Synchronous Reset
  );
```

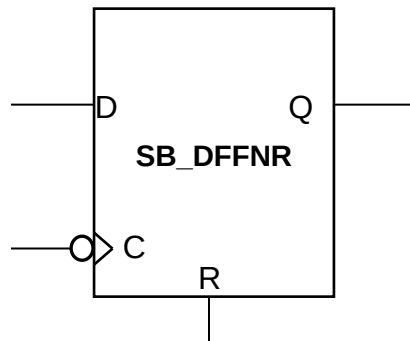
```
-- End of SB_DFFNSR instantiation
```

SB_DFFNR

D Flip-Flop – Negative Edge Clock with Asynchronous Reset

Data: D is loaded into the flip-flop when R is low during the falling clock edge transition.

Reset: R input is active high, overrides all other inputs and asynchronously resets the Q output.



Inputs			Output
R	D	CLK	Q
1	X	X	0
0	0	↘	0
0	1	↘	1
Power on State	X	X	0

Key

↘ Falling Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'

Input C: Logic '0'

Input R: Logic '0'

Verilog Instantiation

// SB_DFFNR - D Flip-Flop – Negative Edge Clock, Reset is asynchronous to the clock.

```
SB_DFFNR SB_DFFNR_inst (  
    .Q(Q),           // Registered Output  
    .C(C),           // Clock  
    .D(D),           // Data  
    .R(R),           // Asynchronously Reset  
);
```

// End of SB_DFFNR instantiation

VHDL Instantiation

-- SB_DFFNR - D Flip-Flop – Negative Edge Clock, Reset is asynchronous to the clock.

```
SB_DFFNR_inst : SB_DFFNR
  port map (
    Q => Q,          -- Registered Output
    C => C,          -- Clock
    D => D,          -- Data
    R => R           -- Asynchronously Reset
  );

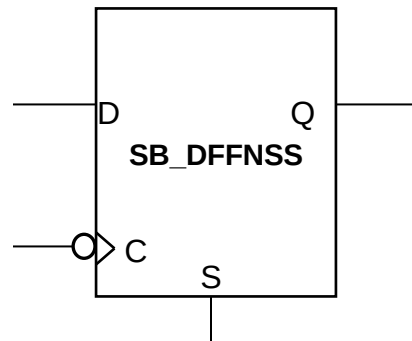
-- End of SB_DFFNR instantiation
```

SB_DFFNSS

D Flip-Flop – Negative Edge Clock with Synchronous Set

Data: D is loaded into the flip-flop when S is low during the falling clock edge transition.

Set: S input is active high, overrides all other inputs and synchronously sets the Q output.



Inputs			Output
S	D	C	Q
1	X	↘	1
0	0	↘	0
0	1	↘	1
Power on State	X	X	0

Key

↘ Falling Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'

Input C: Logic '0'

Input S: Logic '0'

Verilog Instantiation

// SB_DFFNSS - D Flip-Flop – Negative Edge Clock, Set is synchronous with the falling clock edge,

```
SB_DFFNSS SB_DFFNSS_inst (  
    .Q(Q),           // Registered Output  
    .C(C),           // Clock  
    .D(D),           // Data  
    .S(S)            // Synchronous Set  
);
```

// End of SB_DFFNSS instantiation

VHDL Instantiation

-- SB_DFFNSS - D Flip-Flop – Negative Edge Clock, Set is synchronous with the falling clock edge,

```
SB_DFFNSS_inst : SB_DFFNSS
port map (
  Q => Q,          -- Registered Output
  C => C,          -- Clock
  D => D,          -- Data
  S => S           -- Synchronous Set
);
```

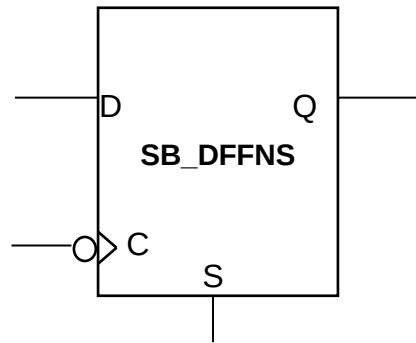
-- End of SB_DFFNSS instantiation

SB_DFFNS

D Flip-Flop – Negative Edge Clock with Asynchronous Set

Data: D is loaded into the flip-flop when S is low during the falling clock edge transition.

Set: S input is active high, overrides all other inputs and asynchronously sets the Q output.



Inputs			Output
S	D	C	Q
1	X	X	1
0	0	↘	0
0	1	↘	1
Power on State	X	X	0

Key

↘ Falling Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'

Input C: Logic '0'

Input S: Logic '0'

Verilog Instantiation

// SB_DFFNS - D Flip-Flop – Negative Edge Clock, Set is asynchronous to the falling clock edge,

```
SB_DFFNS    SB_DFFNS_inst (
    .Q(Q),           // Registered Output
    .C(C),           // Clock
    .D(D),           // Data
    .S(S),           // Asynchronous Set
);
```

// End of SB_DFFNS instantiation

VHDL Instantiation

-- SB_DFFNS - D Flip-Flop – Negative Edge Clock, Set is asynchronous to the falling clock edge

```
SB_DFFNS_inst : SB_DFFNS
  port map (
    Q => Q,          -- Registered Output
    C => C,          -- Clock
    D => D,          -- Data
    S => S           -- Asynchronous Set
  );

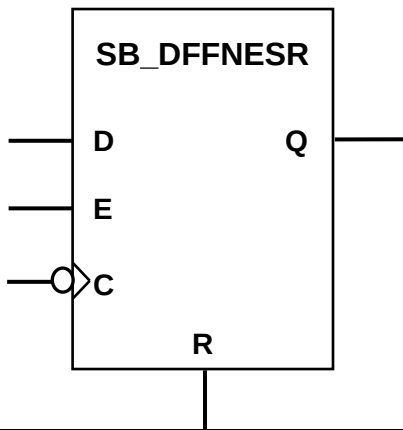
-- End of SB_DFFNS instantiation
```


SB_DFFNESR

D Flip-Flop – Negative Edge Clock, Enable and Synchronous Reset

Data: D is loaded into the flip-flop when R is low and E is high during the falling clock edge transition.

Reset: Asserting R when the Clock Enable E is high, synchronously resets the Q output during the falling clock edge.



Inputs				Output
R	E	D	C	Q
1	1	X	↘	0
X	0	X	X	Previous Q
0	1	0	↘	0
0	1	1	↘	1
Power on State	X	X	X	0

Key
↘ Falling Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'
Input C: Logic '0'
Input R: Logic '0'
Input E: Logic '1'

Note that explicitly connecting a Logic '1' value to port E will result in a non-optimal implementation, since an extra LUT will be used to generate the Logic '1'. If the user's intention is to keep the FF always enabled, it is recommended that either port E be left unconnected, or the corresponding FF without a Clock Enable port be used.

Verilog Instantiation

```
// SB_DFFNESR - D Flip-Flop – Negative Edge Clock, Reset is synchronous with falling clock edge Clock Enable.
```

```

SB_DFFNCSR    SB_DFFNCSR_inst (
    .Q(Q),      // Registered Output
    .C(C),      // Clock
    .E(E),      // Clock Enable
    .D(D),      // Data
    .R(R)       // Synchronous Reset
);

```

```

// End of SB_DFFNCSR instantiation

```

VHDL Instantiation

-- SB_DFFNCSR - D Flip-Flop – Negative Edge Clock, Reset is synchronous with falling clock edge Clock Enable.

```

SB_DFFNCSR_inst : SB_DFFNCSR
    port map (
        Q => Q,      -- Registered Output
        C => C,      -- Clock
        E => E,      -- Clock Enable
        D => D,      -- Data
        R => R       -- Synchronous Reset
    );

```

```

-- End of SB_DFFNCSR instantiation

```